

Year 10 Engineering Technology - Wind Powered USB Phone Charger

20-week scope and sequence for the Wind USB Charger / Alternate Energy unit

Unit Overview

Students design, construct, test and evaluate a compact wind-powered USB charging prototype. The unit connects alternative energy theory, mechanical energy transfer, technical graphics, practical construction, safe low-voltage testing and evidence-based evaluation.

This document spreads the unit across a full 20-week semester. Timing can still be adjusted around excursions, assessment windows and school events.

Stage 5 Outcomes

Outcome	Official outcome wording used for this unit
EGT5-SAF-01	applies risk management and safe work practices in engineering contexts
EGT5-COM-01	communicates ideas, concepts and solutions for engineering practice
EGT5-EVL-01	investigates and evaluates engineering systems and solutions
EGT5-IVT-01	explains engineering practices and the influence of technologies used in engineering industries
EGT5-GRP-01	develops and applies technical graphics in engineering contexts
EGT5-MEA-01	applies mechanical analysis and practical testing to investigate engineering concepts
EGT5-USE-01	selects and applies materials for engineering projects
EGT5-ENV-01	analyses relationships between engineering design, production and sustainability

Content Alignment

Content area	Where it appears across the 20 weeks
Alternative energy context	Weeks 1-4 through renewable/non-renewable comparison, wind generation, storage, cost and impacts.
Mechanical energy transfer	Weeks 2, 6-8 and 14-18 through impellor, shaft, bearing, gear/cog and alignment testing.

Safe work practices	Weeks 1, 5, 8 and 13-16 through rotating-part guarding, tool use and low-voltage output testing.
Engineering design process	Weeks 5-12 through brief analysis, constraints, concept development, final design and build sequence.
Materials and components	Weeks 6, 7, 11, 13 and 19 through blade, frame, shaft support, bearing and housing decisions.
Testing and troubleshooting	Weeks 8, 15-18 through fair testing, output measurement, fault logging and controlled improvements.
Technical communication	Weeks 9-12 and 18-20 through sketches, CAD/drawings, test tables, folio and evaluation.
Sustainability and impact	Weeks 3, 4 and 19 through material lifecycle, recyclability, portability and real-world energy use.

20-Week Scope and Sequence

Week	Focus	Learning experiences	Target outcomes	Evidence
1	Unit launch and alternative energy context	Renewable and non-renewable energy; project context; WHS for rotating parts and low-voltage testing.	EGT5-SAF-01, EGT5-IVT-01, EGT5-COM-01	Safety notes, energy comparison and project brief response.
2	Wind turbine operation	Energy transfer chain; turbine parts; generator output; charging and storage limits.	EGT5-IVT-01, EGT5-MEA-01	Energy pathway diagram and vocabulary notes.
3	Renewable energy advantages and limits	Intermittency, storage, charging infrastructure, regional use and reliability issues.	EGT5-ENV-01, EGT5-EVL-01	Advantages/limitations table and short response.
4	Energy costing and lifecycle impact	Cost of energy production; materials, recyclability, disposal and sustainability impacts.	EGT5-ENV-01, EGT5-IVT-01	Research notes and sustainability evidence.
5	Project brief and generator constraints	USB generator unit, dimensions, box size limit, portability, compactness and safety risks.	EGT5-SAF-01, EGT5-COM-01	Constraint sheet and risk list.
6	Impellor research	Blade material, blade angle, number of blades, wind-catcher options and efficiency variables.	EGT5-MEA-01, EGT5-USE-01	Impellor research notes and labelled blade ideas.

7	Mechanical transfer options	Direct drive, gears/cogs, shafts, bearings, friction, alignment and rotation losses.	EGT5-MEA-01, EGT5-EVL-01	Mechanical option comparison and annotated sketch.
8	Test variables and design constraints	Wind source, output measurement, fair testing, guarded moving parts and teacher checkpoint.	EGT5-SAF-01, EGT5-EVL-01	Test-variable plan and teacher check record.
9	Concept sketching	Frame, impellor, shaft support, generator location, USB module placement and housing.	EGT5-GRP-01, EGT5-COM-01	Two or more concept sketches.
10	Concept comparison	Compare concepts using function, safety, compactness, efficiency, build quality and appearance.	EGT5-EVL-01, EGT5-COM-01	Concept comparison table and selected direction.
11	Final design selection and build planning	Justification, materials list, tools, construction sequence and risk-control update.	EGT5-GRP-01, EGT5-SAF-01	Final design justification and build sequence.
12	Final drawing or CAD model	Onshape or final drawing with dimensions, labels, assembly thinking and component locations.	EGT5-GRP-01, EGT5-COM-01	Final drawing/CAD evidence.
13	Frame and shaft-support construction	Begin construction; frame accuracy; shaft support; generator location; guarded moving parts.	EGT5-SAF-01, EGT5-USE-01	Build log and construction photos.
14	Impellor attachment and mechanical alignment	Spin test; bearing support; wobble reduction; friction reduction and quality checks.	EGT5-MEA-01, EGT5-EVL-01	Spin-test notes and mechanical fault log.
15	Electrical connection and safe output testing	Connect generator/USB unit; teacher-check wiring; measure output with approved equipment.	EGT5-SAF-01, EGT5-MEA-01	Output test data and teacher check record.
16	Fault-finding and construction evidence	Identify low output, poor alignment, unstable frame or blade problems; document fixes.	EGT5-EVL-01, EGT5-COM-01	Fault log, photos and improvement decisions.
17	Controlled improvements	Change one variable: blade angle, drive ratio, alignment, frame stability or wind source.	EGT5-EVL-01, EGT5-MEA-01	Improvement plan and modified prototype evidence.
18	Retesting and performance comparison	Retest improvements, compare results and complete final output testing.	EGT5-EVL-01, EGT5-COM-01	Before/after data and final test table.

19	Sustainability and portability evaluation	Review portability, size limit, materials, waste, construction quality and real-world use.	EGT5-ENV-01, EGT5-EVL-01	Evaluation draft using test evidence.
20	Folio completion and submission	Final testing photos, folio completion, final evaluation and submission checkpoint.	EGT5-COM-01, EGT5-EVL-01, EGT5-IVT-01	Completed folio/report and project reflection.

Assessment and Evidence

- Prototype: a compact wind-powered USB charger or generator model that demonstrates wind capture, mechanical transfer and output testing.
- Folio/report: research, constraints, concept sketches, final design choice, drawing/CAD, build log, test data, improvements and final evaluation.
- Teacher observation: safe tool use, guarded rotating parts, low-voltage testing, collaboration and workshop conduct.
- Evaluation: output performance, reliability, portability, sustainability, construction quality and justified improvements.